START

7th International Conference on Bacteriophage Research and Antimicrobial Resistance

Lytic Legends Edition · 13 rounds of plaques, probes and punchlines

Teams

Lytic Legends

Dr. Ruma

Ms. Priya

Mr. Karan

Biofilm Breakers

Dr. Rupali

Dr. Prachi

Mr. Prajasattak

Phage Pharmers

Team member 1

Team member 2

Team member 3

Virion Architects

Team member 1

Team member 2

Team member 3

Rounds

1. GENERAL QUESTIONS
2. HISTORY OF BACTERIOPHAGES
3. PHAGE RELATED PROTOCOLS
4. PHAGE INFORMATICS
5. PHAGE IDENTIFICATION
6. IDENTIFY THE FIGURE
7. PHAGE THERAPY & CLINICAL FRONTIERS NEW
8. PHAGE ACRONYMS
9. PHAGE ECOLOGY & ONE HEALTH NEW
10. BACTERIAL DEFENSES & PHAGE COUNTER-DE

BREAK (5 min)

11. RAPID FIRE
12. TIE-BREAKER ROUND
13. EMOJI REBUS & PHAGE PERSONALS NEW

Markings

Rounds 1-5	+10 / 0	30 sec
Rounds 6-10	+10 / -5	30 sec
Rapid Fire	+5 / 0, no passing	60 sec / team
Tie-Breaker	+20 / -10, passed +5	60 sec
Emoji Rebus & Lytic Laughs	+10 / 0	30 sec

Each MCQ round: 8 team questions + 2 audience questions

1

GENERAL QUESTIONS

Correct +10 | Wrong 0 | 30 sec

ROUND 1

Phage genomes are unusually compact, with overlapping genes and non-canonical start codons. The best explanation is

A. Selection for genome compaction under capsid packaging constraints

B. A high mutation rate that erases intergenic regions

C. Limitation of host ribosomes during infection

D. An artifact of automated annotation pipelines

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ROUND 1

The ~50·60 atm internal pressure that drives genome ejection from tailed phage capsids is generated mainly by

A. The host proton-motive force pulling DNA inward

B. Dense inverse-spool DNA packing with electrostatic self-repulsion

C. ATP hydrolysis by the portal motor during ejection

D. Osmotic water influx upon receptor binding

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ROUND 1

At an MOI of exactly 1, the fraction of host cells receiving at least one phage (Poisson distribution) is approximately

A 37%

B 50%

C 63%

D 90%

ROUND 1

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In the lambda lysis cassette (S, R, Rz, Rz1), the spanins Rz/Rz1 act by

A Degrading peptidoglycan from the periplasm

B Timing holin insertion into the inner membrane

C Recycling murein for the next infection cycle

D Fusing inner and outer membranes after endolysin action

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Optimality models of lysis timing predict that the evolutionarily stable latent period balances

A. Burst size against the number of lysis cycles completed per unit time

B. Adsorption rate against the eclipse period

C. Genome size against capsid interior volume

D. Latent period against the multiplicity of infection

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ROUND 1

Botstein's ·modular theory· of phage evolution proposes that

A Phage genes evolve strictly vertically with their hosts

B Functional modules (head, tail, lysis, lysogeny) recombine across phage lineages

C All tailed phages preserve a single ancestral gene order

D Phage genomes change only by point mutation

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ROUND 1

Jumbo phages such as ϕKZ protect their replicating DNA inside a compartment whose shell is built from

A Host-derived lipid bilayer

B Condensed RNA and polyamines

C The protein chimallin (a proteinaceous phage nucleus)

D Crystallized terminase

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The kinetics of phage adsorption to susceptible hosts are best described as

A Zero-order until receptors saturate

B First-order in phage only, independent of host density

C Logistic, limited by burst size

D Second-order overall: first-order in both phage and host concentration

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ROUND 1

AUDIENCE

Which is generally SMALLER?

A A bacteriophage

B A bacterium

C A yeast cell

D A red blood cell

ROUND 1

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B A bacterium

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ROUND 1

AUDIENCE

Phage contamination of starter cultures is a classic industrial problem in which fermentation?

A Bread baking

B Cheese and yogurt making

C Beer brewing

D Wine making

ROUND 1

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2

HISTORY OF BACTERIOPHAGES

Correct +10 | Wrong 0 | 30 sec

ROUND 2

In the Hershey·Chase experiment, what physically separated ·phage ghosts· from infected bacteria?

A Shearing in a Waring blendor

B Differential centrifugation of isotopes

C Freeze·thaw lysis of the cells

D Ethanol precipitation of free phage

ROUND 2

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ROUND 2

Benzer's rII recombination mapping was only possible because rII mutants fail to plaque on

A Escherichia coli B

B Salmonella Typhimurium LT2

C Escherichia coli K12(·)

D Pseudomonas aeruginosa PAO1

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Crick, Brenner and colleagues (1961) proved the triplet nature of the code using proflavin-induced frameshifts in

A The T4 rII gene

B Lambda cI

C MS2 coat protein

D T7 gene 10

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Weigle reactivation (1960) showed that UV-damaged lambda is rescued by

A. Recombination with co-infecting intact phage

B. SOS-induced host repair functions

C. Photoreactivation of phage RNA

D. Mitomycin C induction of the prophage

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ROUND 2

·Host-controlled variation· of lambda grown on E. coli K-12 versus B directly led to the discovery of

A CRISPR-Cas immunity

B Abortive infection systems

C Restriction·modification systems

D Superinfection exclusion

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ROUND 2

The Luria·Delbrück fluctuation test established that phage-resistance mutations

A Are instructed by phage exposure

B Spread by transduction

C Require SOS induction

D Arise spontaneously before phage exposure

ROUND 2

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A Are instructed by phage exposure

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D Arise spontaneously before phage exposure

ROUND 2

Frederick Twort's 1915 filterable agent was first observed lysing cultures of

A Shigella dysenteriae

B Vibrio cholerae

C Bacillus anthracis

D Staphylococcus

ROUND 2

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ROUND 2

Which 1930s assessment most accelerated the decline of commercial phage therapy in the United States?

A The Flexner Report

B The Eaton & Bayne-Jones review of phage preparations

C The Nuremberg Code

D A Lister Institute ban

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AUDIENCE

The 2017 compassionate-use case that brought phage therapy to global headlines treated a patient with

A Disseminated Acinetobacter baumannii infection

B MRSA endocarditis

C XDR tuberculosis

D Candida auris sepsis

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The 1969 Nobel Prize (Delbrück, Hershey, Luria) honoured work on the replication and genetic structure of

A Influenza virus

B Bacteriophages

C Poliovirus

D Retroviruses

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3

PHAGE RELATED PROTOCOLS

Correct +10 | Wrong 0 | 30 sec

ROUND 3

Top-agar concentrations above ~0.7% shrink plaque sizes mainly because they

A Melt at lower temperatures

B Select for resistant mutants

C Chelate Ca2+ and Mg2+

D Restrict phage diffusion through the matrix

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ROUND 3

The particle-to-PFU ratio of a lysate is obtained by combining

A qPCR with PFGE

B CsCl gradients with SYBR Gold

C Flow cytometry (particle counts) with plaque assay

D LAL with TEM

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ROUND 3

During phage DNA extraction, contaminating free bacterial DNA is removed selectively by

A DNase I treatment before capsid lysis (capsid DNA is protected)

B Proteinase K before lysis

C Ethanol precipitation alone

D RNase A after lysis

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ROUND 3

In a CsCl isopycnic gradient, phage bands at ~1.5 g/cm³. A lighter band at ~1.3 g/cm³ is most likely

A Protein and membrane vesicles

B Host chromosomal DNA

C Intact ribosomes

D Free exopolysaccharide

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ROUND 3

Chloroform clarification of a lysate must NOT be used for

A Microviruses

B Enveloped phages such as ·6

C Filamentous inoviruses

D Tailed dsDNA phages

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ROUND 3

In the Ellis·Delbrück one-step growth experiment, unadsorbed phage is neutralized by

A Using stationary-phase hosts

B Anti-phage serum (or extreme dilution)

C Chilling to 4 °C

D Adding chloroform at time zero

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ROUND 3

A spot test giving confluent lysis that disappears on dilution, without plaques, indicates

A Lysis from without at high phage input

B A fully permissive host

C Lysogenic conversion

D Capsular blocking of adsorption

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A spot test giving confluent lysis that disappears on dilution, without plaques, indicates

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ROUND 3

For phage stocks kept at 4 °C, which additive limits titer loss by adsorption to tube walls?

A 100 mM EDTA

B 50% glycerol

C Sodium azide

D Gelatin (or BSA)

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AUDIENCE

In a spot test, a clear zone where phage suspension was dropped on a bacterial lawn indicates

A Medium contamination

B The strain is sensitive (lysis)

C The strain is resistant

D A mutation arose

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ROUND 3

AUDIENCE

In a plaque assay, each plaque originates from

A A free DNA fragment

B An antibiotic disc

C A single infectious phage particle

D A single bacterial cell

ROUND 3

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4

PHAGE INFORMATICS

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ROUND 4

Under MIMAG standards, a high-quality draft viral genome requires

A ·90% completeness, <5% contamination, plus tRNAs and identified termini

B Any contig >10 kb carrying a capsid gene

C 100× coverage regardless of completeness

D Presence of an integrase gene

ROUND 4

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ROUND 4

In read-mapping to a phage reference, terminal redundancy of the genome appears as

A. A coverage gap at the cos site

B. Elevated coverage across both genome ends

C. Uniform coverage with spikes at tRNA genes

D. Coverage falling to zero at the ends

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ROUND 4

Anti-CRISPR (acr) genes in prophages are typically found next to

A Host CRISPR arrays

B tRNA clusters

C aca genes encoding HTH-type transcriptional regulators

D Direct terminal repeats

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ROUND 4

·Viral tagging· (fluorescently labelled phage + flow sorting of single cells) is used to

A Measure buoyant density

B Cluster genomes by tetranucleotide frequency

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D Link phages to their hosts at single-cell resolution

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ROUND 4

VIRIDIC demarcates viral species and genera using

A Intergenomic nucleotide similarity

B Shared tRNA counts

C Codon-usage distance

D Capsid morphology vectors

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ROUND 4

CheckV evaluates viral metagenome-assembled genomes for

A Host range prediction

B Completeness, contamination and provirus boundaries

C Plaque morphology class

D AMR gene content

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ROUND 4

PhageTerm determines

A Lysis timing from growth curves

B Taxonomic family from marker genes

C Genome termini and the packaging mechanism (cos, pac, headful)

D Optimal cocktail composition

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ROUND 4

Comparing phage codon adaptation to candidate hosts helps predict

A Burst size

B Capsid diameter

C Genome termini type

D The likely host / recent host adaptation

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ROUND 4

AUDIENCE

The complete collection of viruses in a sample is called the

A Microbiome

B Virome

C Proteome

D Plasmidome

ROUND 4

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AUDIENCE

A prophage region whose GC content differs strongly from the rest of the bacterial chromosome suggests

A Recently acquired foreign DNA

B A better plaque morphology

C A smaller capsid

D An RNA genome

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PHAGE IDENTIFICATION

Correct +10 | Wrong 0 | 30 sec

ROUND 5

Under current ICTV taxonomy, phage T7 (formerly Podoviridae) is placed in the family

A Autographiviridae

B Ackermannviridae

C Drexlerviridae

D Herelleviridae

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The lipid envelope of cystovirus ·6 is

A Assembled from host-derived membrane lipids

B Acquired from outer-membrane vesicles of other species

C A protein-only pseudo-envelope

D Synthesized de novo with phage-made sterols

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In levivirus MS2, the single-copy maturation (A) protein

A Triggers lysis

B Forms the icosahedral capsid

C Binds the F-pilus and escorts the RNA into the cell

D Acts as the replicase subunit

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ROUND 5

The portal vertex of tailed phages is typically a ring of

A 5 subunits

B ~12·13 portal subunits

C 60 subunits matching T=1 symmetry

D 1 subunit

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Transcription of jumbo phage ·KZ is independent of the host RNA polymerase because the phage

A Transcribes via a ribozyme

B Hijacks host Pol II

C Uses a phage-encoded sigma factor with host RNAP

D Encodes its own multisubunit RNA polymerase

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Filamentous inoviruses leave the host without lysis by

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C Budding with a lipid envelope

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T4 whiskers (wac/fib3) function to

A. Anchor the baseplate to the capsid

B. Hold long tail fibers folded against the tail before adsorption

C. Contract and pierce the outer membrane

D. Package DNA into the procapsid

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ROUND 5

The collar of phage T4 forms

A The portal·terminase bridge

B The head·tail junction

C The sheath·tube interface

D The fiber·baseplate contact

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AUDIENCE

Under TEM, a tailed phage with a small head and a very SHORT non-contractile tail is a

A Siphovirus

B Filamentous phage

C Podovirus

D Myovirus

ROUND 5

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AUDIENCE

Turbid (cloudy) plaques usually suggest

A A temperate phage forming lysogens

B A strictly lytic phage

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6

IDENTIFY THE FIGURE

Correct +10 | Wrong -5 | 30 sec

ROUND 6

In the arrowed interval of this one-step growth curve, no intracellular infectious virions can be recovered. This portion is the

A Eclipse period

B Rise period

C Burst phase

D Adsorption plateau

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ROUND 6

The clear HALO extending beyond the plaque edge in this plate is most likely due to

A An antibiotic in the medium

B Extracellular depolymerase degrading the capsule/EPS

C Lysis from without

D Phage-resistant mutants

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ROUND 6

The arrowed long tail fibers bind reversibly. Irreversible adsorption of T4 is then mediated by

A. Short tail fibers engaging the inner LPS core after baseplate triggering

B. A second long-fiber receptor

C. Teichoic acid bridging

D. Pilus retraction

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ROUND 6

The integrated state depicted here is maintained by

A Cro

B The cI repressor

C CII

D N antitermination

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The integrated state depicted here is maintained by

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ROUND 6

Top agar is cooled to ~45 °C before mixing because this

A Increases plaque turbidity

B Speeds up lawn growth

C Prevents Ca2+ precipitation

D Keeps agar liquid while sparing phage and host from heat

ROUND 6

Top agar is cooled to ~45 °C before mixing because this

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D Keeps agar liquid while sparing phage and host from heat

ROUND 6

The arrowed band at ~1.5 g/cm³ holds phage. A fainter band at ~1.3 g/cm³ in the same tube would be

A Protein and membrane vesicles

B Host rRNA

C Intact progeny virions

D DNA·protein concatemers

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ROUND 6

Unlike virion 1, the MIDDLE virion injects its DNA by

A Contracting its tail more slowly

B Injecting RNA instead of DNA

C Receptor-triggered conformational change without sheath contraction

D Consuming host ATP for contraction

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ROUND 6

To penetrate the matrix shown in this biofilm diagram, the most useful phage-encoded enzymes are

A DNA ligases

B Integrases

C RNA polymerases

D Depolymerases

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ROUND 6

On this spot-assay plate, a drop giving confluent clearing but no plaques on dilution indicates

A. A fully permissive host

B. Lysogenic conversion

C. Lysis from without / killing without productive infection

D. Depolymerase secretion

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ROUND 6

Plate A shows clear plaques, plate B turbid plaques. The turbidity in plate B is caused by

A Incomplete top-agar setting

B Growth of immune lysogens within the plaque

C Contaminating fungi

D A very small burst size

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7

PHAGE THERAPY & CLINICAL FRONTIERS

Correct +10 | Wrong -5 | 30 sec

ROUND 7

NEW TOPIC

The 2019 Nature Medicine report of an engineered three-phage cocktail successfully treated which infection in a cystic fibrosis patient?

A Staphylococcus aureus osteomyelitis

B Disseminated Mycobacterium abscessus infection

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D Acinetobacter baumannii sepsis

ROUND 7

NEW TOPIC

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ROUND 7

NEW TOPIC

Since 2018, Belgium delivers therapeutic phages to patients as

A Over-the-counter medicines

B Vaccines

C Medical devices

D Active pharmaceutical ingredients within magistral preparations

ROUND 7

NEW TOPIC

Since 2018, Belgium delivers therapeutic phages to patients as

A Over-the-counter medicines

B Vaccines

C Medical devices

D Active pharmaceutical ingredients within magistral preparations

ROUND 7

NEW TOPIC

The PhagoBurn trial (Lancet ID, 2018) evaluated phage therapy against which pathogen in burn wounds?

A Escherichia coli

B Staphylococcus aureus

C Pseudomonas aeruginosa

D Klebsiella pneumoniae

ROUND 7

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ROUND 7

NEW TOPIC

LBP-EC01, the first engineered (CRISPR-Cas3-enhanced) phage tested in a randomized placebo-controlled trial, targets

A Escherichia coli in urinary tract infection

B Clostridioides difficile colitis

C Pseudomonas aeruginosa in CF

D Staphylococcus aureus bacteremia

ROUND 7

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ROUND 7

NEW TOPIC

·Phage steering· mitigates AMR by

A Increasing the MOI until resistance collapses

B Selecting phage-resistant mutants whose fitness cost re-sensitizes them to antibiotics

C Blocking phage receptors with antibodies

D Using phages to inactivate antibiotics

ROUND 7

NEW TOPIC

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A Increasing the MOI until resistance collapses

B Selecting phage-resistant mutants whose fitness cost re-sensitizes them to antibiotics

C Blocking phage receptors with antibodies

D Using phages to inactivate antibiotics

ROUND 7

NEW TOPIC

A key immunological limitation of repeated intravenous phage dosing is

A Permanent T-cell exhaustion

B Neutralizing antibodies that accelerate phage clearance

C Phages cannot enter the bloodstream

D Immediate IgE anaphylaxis in every patient

ROUND 7

NEW TOPIC

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B Neutralizing antibodies that accelerate phage clearance

C Phages cannot enter the bloodstream

D Immediate IgE anaphylaxis in every patient

ROUND 7

NEW TOPIC

In the United States, compassionate-use phage therapy is typically authorized through

A An FDA emergency/expanded-access IND (eIND)

B EPA registration

C An OTC monograph

D A USDA rule

ROUND 7

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ROUND 7

NEW TOPIC

Before clinical use, a therapeutic phage genome must be screened to confirm

A Absence of toxin, AMR and virulence genes

B GC content above 60%

C Uracil instead of thymine

D Presence of an integrase gene

ROUND 7

NEW TOPIC

Before clinical use, a therapeutic phage genome must be screened to confirm

A Absence of toxin, AMR and virulence genes

B GC content above 60%

C Uracil instead of thymine

D Presence of an integrase gene

ROUND 7

NEW TOPIC

AUDIENCE

Commercial phage preparations sprayed on ready-to-eat food to control Listeria are an example of

A Pasteurization

B Vaccination

C Fermentation

D Biocontrol

ROUND 7

NEW TOPIC

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Commercial phage preparations sprayed on ready-to-eat food to control Listeria are an example of

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B Vaccination

C Fermentation

D Biocontrol

ROUND 7

NEW TOPIC

AUDIENCE

Compared with antibiotics, a distinctive advantage of phages is that they

A Kill human cells selectively

B Never encounter resistance

C Self-amplify at the site of infection

D Work without any bacteria present

ROUND 7

NEW TOPIC

AUDIENCE

Compared with antibiotics, a distinctive advantage of phages is that they

A Kill human cells selectively

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8

PHAGE ACRONYMS

Correct +10

Wrong -5

30 sec

ROUND 8

In phage anti-CRISPR biology, Aca proteins are

A Acr-associated HTH transcriptional regulators of acr operons

B Accessory capsid antigens

C Acid-stable coat adapters

D attC-associated recombinases

ROUND 8

In phage anti-CRISPR biology, Aca proteins are

A Acr-associated HTH transcriptional regulators of acr operons

B Accessory capsid antigens

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D attC-associated recombinases

ROUND 8

In phage genomics, DTRs and ITRs refer to

A Degenerate tandem recombinases

B Direct and inverted terminal repeats

C Density-transfer ratios

D Double-stranded transcriptional regions

ROUND 8

In phage genomics, DTRs and ITRs refer to

A Degenerate tandem recombinases

B Direct and inverted terminal repeats

C Density-transfer ratios

D Double-stranded transcriptional regions

ROUND 8

In the name vB_EcoM-VPAE1, the ·vB· prefix denotes

A Virus of Bacteria (followed by host genus and morphotype)

B Verified bacteriophage

C Viral barcode

D Vector backbone

ROUND 8

In the name vB_EcoM-VPAE1, the ·vB· prefix denotes

A Virus of Bacteria (followed by host genus and morphotype)

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C Viral barcode

D Vector backbone

ROUND 8

MIMAG stands for

A Metagenome Integration Map

B Minimum Information about Microbial Archaeal Genomes

C Minimum Information about a Metagenome-Assembled Genome

D Microbial Image Magnification

ROUND 8

MIMAG stands for

A Metagenome Integration Map

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ROUND 8

In phage genomics, AMGs are

A Auxiliary metabolic genes

B Amplified mobile genes

C Ancestral marker genes

D Adaptive mutation groups

ROUND 8

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B Amplified mobile genes

C Ancestral marker genes

D Adaptive mutation groups

ROUND 8

PHROGs are

A Prophage host-range groups

B Phage restriction operon groups

C Plaque heterogeneity ratings

D Phage orthologous groups for functional annotation

ROUND 8

PHROGs are

A Prophage host-range groups

B Phage restriction operon groups

C Plaque heterogeneity ratings

D Phage orthologous groups for functional annotation

ROUND 8

In phage·mucus research, the BAM model stands for

A Bacterial adhesion matrix

B Bacteriophage adherence to mucus

C Biofilm-associated microcolony

D Buffered agar medium

ROUND 8

In phage·mucus research, the BAM model stands for

A Bacterial adhesion matrix

B Bacteriophage adherence to mucus

C Biofilm-associated microcolony

D Buffered agar medium

ROUND 8

The LAL assay reagent used in phage formulation is derived from

A Lactic acid bacteria

B Lysed adenovirus libraries

C Lipid antigen liposomes

D Limulus (horseshoe crab) amebocyte lysate

ROUND 8

The LAL assay reagent used in phage formulation is derived from

A Lactic acid bacteria

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D Limulus (horseshoe crab) amebocyte lysate

ROUND 8

AUDIENCE

In One-Health discussions, AMR stands for

A Acquired mobile resistance

B Adaptive mutation rate

C Antimicrobial resistance

D Antibiotic modifier region

ROUND 8

AUDIENCE

In One-Health discussions, AMR stands for

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B Adaptive mutation rate

C Antimicrobial resistance

D Antibiotic modifier region

ROUND 8

AUDIENCE

In the phage morphology lab, TEM stands for

A Thermal emission microscopy

B Transmission electron microscopy

C Total evanescent measurement

D Tunneling electron mapping

ROUND 8

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In the phage morphology lab, TEM stands for

A Thermal emission microscopy

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9

PHAGE ECOLOGY & ONE HEALTH

Correct +10 | Wrong -5 | 30 sec

ROUND 9

NEW TOPIC

The ·viral shunt· refers to

A Phage lysis redirecting microbial biomass into dissolved organic matter

B Aerosolization of virions

C Viral transport in the bloodstream

D Phage migration through soil

ROUND 9

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A Phage lysis redirecting microbial biomass into dissolved organic matter

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D Phage migration through soil

ROUND 9

NEW TOPIC

A millilitre of surface seawater typically contains about

A 10^3 phages

B 10^12 phages

C 10^7 phages

D 10^15 phages

ROUND 9

NEW TOPIC

A millilitre of surface seawater typically contains about

A 10^3 phages

B 10^12 phages

C 10^7 phages

D 10^15 phages

ROUND 9

NEW TOPIC

Cyanophages infecting Prochlorococcus often carry which host-derived gene to sustain photosynthesis during infection?

A nifH

B psbA (photosystem II D1 protein)

C rpoB

D atpD

ROUND 9

NEW TOPIC

Cyanophages infecting Prochlorococcus often carry which host-derived gene to sustain photosynthesis during infection?

A nifH

B psbA (photosystem II D1 protein)

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D atpD

ROUND 9

NEW TOPIC

·Kill-the-winner· dynamics predicts that

A Rare bacterial species are lysed first

B The most abundant host is preferentially hit by lytic phages

C Phages prefer dormant hosts

D Lysogeny dominates in rich environments

ROUND 9

NEW TOPIC

·Kill-the-winner· dynamics predicts that

A Rare bacterial species are lysed first

B The most abundant host is preferentially hit by lytic phages

C Phages prefer dormant hosts

D Lysogeny dominates in rich environments

ROUND 9

NEW TOPIC

Which phage is among the most prevalent human-associated viruses on Earth, carried in the gut of a large fraction of adults?

A Phage ·6

B Phage lambda

C crAssphage

D Phage T4

ROUND 9

NEW TOPIC

Which phage is among the most prevalent human-associated viruses on Earth, carried in the gut of a large fraction of adults?

A Phage ·6

B Phage lambda

C crAssphage

D Phage T4

ROUND 9

NEW TOPIC

The cholera toxin genes (ctxAB) are encoded by

A Phage P1

B Phage lambda

C Phage Mu

D Phage CTX·

ROUND 9

NEW TOPIC

The cholera toxin genes (ctxAB) are encoded by

A Phage P1

B Phage lambda

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D Phage CTX·

ROUND 9

NEW TOPIC

Diphtheria toxin is encoded by

A Phage CTX·

B The ·-prophage (corynephage ·)

C Phage P22

D Phage ·KZ

ROUND 9

NEW TOPIC

Diphtheria toxin is encoded by

A Phage CTX·

B The ·-prophage (corynephage ·)

C Phage P22

D Phage ·KZ

ROUND 9

NEW TOPIC

Shiga toxin genes (stx) carried by lambdoid prophages are central to the virulence of

A Shiga-toxigenic E. coli (STEC)

B Salmonella Typhi

C Listeria monocytogenes

D Campylobacter jejuni

ROUND 9

NEW TOPIC

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B Salmonella Typhi

C Listeria monocytogenes

D Campylobacter jejuni

ROUND 9

NEW TOPIC

AUDIENCE

Municipal sewage is extremely rich in phages mainly because

A High bacterial densities support massive phage replication

B Phages resist freezing

C UV light creates virions

D Chlorine selects for viruses

ROUND 9

NEW TOPIC

AUDIENCE

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A High bacterial densities support massive phage replication

B Phages resist freezing

C UV light creates virions

D Chlorine selects for viruses

ROUND 9

NEW TOPIC

AUDIENCE

Phages are part of ·One Health· thinking because they

A Replace all vaccines

B Chemically purify water

C Only infect humans

D Connect human, animal and environmental microbiomes, including AMR dynamics

ROUND 9

NEW TOPIC

AUDIENCE

Phages are part of ·One Health· thinking because they

A Replace all vaccines

B Chemically purify water

C Only infect humans

D Connect human, animal and environmental microbiomes, including AMR dynamics

10

BACTERIAL DEFENSES & PHAGE COUNTER-DEFENSE

Correct +10 | Wrong -5 | 30 sec

ROUND 10

NEW TOPIC

CRISPR-Cas adaptive immunity stores phage-derived sequences in the bacterial genome as

A Telomeres

B Spacers

C Operators

D Introns

ROUND 10

NEW TOPIC

CRISPR-Cas adaptive immunity stores phage-derived sequences in the bacterial genome as

A Telomeres

B Spacers

C Operators

D Introns

ROUND 10

NEW TOPIC

Anti-CRISPR (Acr) proteins protect phages by

A Blocking phage adsorption

B Degrading bacterial DNA

C Inducing host lysis early

D Inhibiting Cas proteins

ROUND 10

NEW TOPIC

Anti-CRISPR (Acr) proteins protect phages by

A Blocking phage adsorption

B Degrading bacterial DNA

C Inducing host lysis early

D Inhibiting Cas proteins

ROUND 10

NEW TOPIC

Restriction–modification systems defend bacteria by

A Pumping phages out by efflux

B Methylating phage tail fibers

C Cleaving unmethylated foreign DNA

D Blocking bacterial ribosomes

ROUND 10

NEW TOPIC

Restriction·modification systems defend bacteria by

A Pumping phages out by efflux

B Methylating phage tail fibers

C Cleaving unmethylated foreign DNA

D Blocking bacterial ribosomes

ROUND 10

NEW TOPIC

Abortive infection (Abi) systems protect a bacterial population by

A Repairing damaged DNA

B Triggering death or dormancy of the infected cell

C Preventing phage adsorption

D Degrading extracellular phages

ROUND 10

NEW TOPIC

Abortive infection (Abi) systems protect a bacterial population by

A Repairing damaged DNA

B Triggering death or dormancy of the infected cell

C Preventing phage adsorption

D Degrading extracellular phages

ROUND 10

NEW TOPIC

The arbitrium system lets temperate phages

A Repair their DNA after UV

B Sense a peptide cue of host density to choose lysis vs lysogeny

C Adsorb faster to pili

D Resist CRISPR targeting

ROUND 10

NEW TOPIC

The arbitrium system lets temperate phages

A Repair their DNA after UV

B Sense a peptide cue of host density to choose lysis vs lysogeny

C Adsorb faster to pili

D Resist CRISPR targeting

ROUND 10

NEW TOPIC

Phage VP882 eavesdrops on which bacterial signal to time its lysis decision?

A Teichoic acid

B Lipopolysaccharide

C A quorum-sensing autoinducer

D Host ATP levels

ROUND 10

NEW TOPIC

Phage VP882 eavesdrops on which bacterial signal to time its lysis decision?

A Teichoic acid

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D Host ATP levels

ROUND 10

NEW TOPIC

Phage resistance via loss of a surface receptor often costs the bacterium

A Reduced fitness or virulence, sometimes restored antibiotic sensitivity

B Prophage induction

C Immediate lysis

D CRISPR array loss

ROUND 10

NEW TOPIC

Phage resistance via loss of a surface receptor often costs the bacterium

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C Immediate lysis

D CRISPR array loss

ROUND 10

NEW TOPIC

A classic phage counter-defense against restriction systems is

A DNA modification, e.g. glucosylated 5-hydroxymethylcytosine in T4

B Faster adsorption kinetics

C A longer tail

D A larger capsid

ROUND 10

NEW TOPIC

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ROUND 10

NEW TOPIC

AUDIENCE

Superinfection exclusion means that

A. A resident prophage blocks secondary infection by related phages

B. Two phages co-infect one cell

C. Phages exclude antibiotics from the cell

D. Bacteria pump phages out by efflux

ROUND 10

NEW TOPIC

AUDIENCE

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ROUND 10

NEW TOPIC

AUDIENCE

BREX, DISARM, Thoeris and Gabija are examples of

A Plasmid incompatibility groups

B Staining methods

C Phage families

D Bacterial anti-phage defense systems

ROUND 10

NEW TOPIC

AUDIENCE

BREX, DISARM, Thoeris and Gabija are examples of

A Plasmid incompatibility groups

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C Phage families

D Bacterial anti-phage defense systems

Break Time!

5 MIN

11

RAPID FIRE

Correct +5 | Wrong 0 | 60 sec

ROUND 11 - RAPID FIRE

Lytic Legends - question 1 of 8

Enzyme that degrades the peptidoglycan at the end of the lytic cycle

ROUND 11 - RAPID FIRE

Lytic Legends - question 1 of 8

Enzyme that degrades the peptidoglycan at the end of the lytic cycle

Endolysin (lysozyme)

ROUND 11 - RAPID FIRE

Lytic Legends - question 2 of 8

Term for a phage genome integrated in the host chromosome

ROUND 11 - RAPID FIRE

Lytic Legends - question 2 of 8

Term for a phage genome integrated in the host chromosome

Prophage

ROUND 11 - RAPID FIRE

Lytic Legends - question 3 of 8

UV light or mitomycin C triggers this event in a lysogen

ROUND 11 - RAPID FIRE

Lytic Legends - question 3 of 8

UV light or mitomycin C triggers this event in a lysogen

Prophage induction

ROUND 11 - RAPID FIRE

Lytic Legends - question 4 of 8

The protein shell that encloses the phage genome



ROUND 11 - RAPID FIRE

Lytic Legends - question 4 of 8

The protein shell that encloses the phage genome

Capsid

ROUND 11 - RAPID FIRE

Lytic Legends - question 5 of 8

The myovirus structure that contracts during infection



ROUND 11 - RAPID FIRE

Lytic Legends - question 5 of 8

The myovirus structure that contracts during infection

Tail sheath

ROUND 11 - RAPID FIRE

Lytic Legends - question 6 of 8

Number of progeny released per infected cell

ROUND 11 - RAPID FIRE

Lytic Legends - question 6 of 8

Number of progeny released per infected cell

Burst size

ROUND 11 - RAPID FIRE

Lytic Legends - question 7 of 8

Filamentous phage widely used for phage display

ROUND 11 - RAPID FIRE

Lytic Legends - question 7 of 8

Filamentous phage widely used for phage display

M13 (f1/fd)

ROUND 11 - RAPID FIRE

Lytic Legends - question 8 of 8

Ions added to media because many phages need them for adsorption

ROUND 11 - RAPID FIRE

Lytic Legends - question 8 of 8

Ions added to media because many phages need them for adsorption

Divalent cations (Ca2+ / Mg2+)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 1 of 8

Matrix component degraded by phage depolymerases in a biofilm



ROUND 11 - RAPID FIRE

Biofilm Breakers - question 1 of 8

Matrix component degraded by phage depolymerases in a biofilm

Exopolysaccharide (EPS / capsule)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 2 of 8

Phage-mediated transfer of bacterial DNA between cells



ROUND 11 - RAPID FIRE

Biofilm Breakers - question 2 of 8

Phage-mediated transfer of bacterial DNA between cells

Transduction

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 3 of 8

Microscopy that images phages in a near-native frozen-hydrated state

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 3 of 8

Microscopy that images phages in a near-native frozen-hydrated state

Cryo-EM (cryo-TEM)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 4 of 8

The phage life cycle that ends in host lysis



ROUND 11 - RAPID FIRE

Biofilm Breakers - question 4 of 8

The phage life cycle that ends in host lysis

Lytic cycle

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 5 of 8

Extrachromosomal bacterial element that often carries AMR genes

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 5 of 8

Extrachromosomal bacterial element that often carries AMR genes

Plasmid

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 6 of 8

A phage particle that lacks its genome



ROUND 11 - RAPID FIRE

Biofilm Breakers - question 6 of 8

A phage particle that lacks its genome

Empty capsid (ghost)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 7 of 8

The workbench host bacterium of the T-phages

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 7 of 8

The workbench host bacterium of the T-phages

Escherichia coli

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 8 of 8

Ecological model: phages lyse the most abundant host (3 words, ·kill the ··)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 8 of 8

Ecological model: phages lyse the most abundant host (3 words, ·kill the ··)

Kill-the-winner

ROUND 11 - RAPID FIRE

Phage Pharmers - question 1 of 8

US regulatory route for compassionate phage use (3 letters + IND)



ROUND 11 - RAPID FIRE

Phage Pharmers - question 1 of 8

US regulatory route for compassionate phage use (3 letters + IND)

eIND (emergency/expanded-access IND)

ROUND 11 - RAPID FIRE

Phage Pharmers - question 2 of 8

Full name of PEG, used with NaCl to precipitate phages

ROUND 11 - RAPID FIRE

Phage Pharmers - question 2 of 8

Full name of PEG, used with NaCl to precipitate phages

Polyethylene glycol

ROUND 11 - RAPID FIRE

Phage Pharmers - question 3 of 8

Instrument that separates phages by buoyant density



ROUND 11 - RAPID FIRE

Phage Pharmers - question 3 of 8

Instrument that separates phages by buoyant density

(Ultra)centrifuge

ROUND 11 - RAPID FIRE

Phage Pharmers - question 4 of 8

Enzymes that cut DNA at specific sites, first studied in phage biology



ROUND 11 - RAPID FIRE

Phage Pharmers - question 4 of 8

Enzymes that cut DNA at specific sites, first studied in phage biology

Restriction endonucleases

ROUND 11 - RAPID FIRE

Phage Pharmers - question 5 of 8

Coat proteins used to display peptides on M13 (two names)

ROUND 11 - RAPID FIRE

Phage Pharmers - question 5 of 8

Coat proteins used to display peptides on M13 (two names)

pIII / pVIII

ROUND 11 - RAPID FIRE

Phage Pharmers - question 6 of 8

Route used to deliver phages to CF lungs in the 2025 first-in-human randomized trial

ROUND 11 - RAPID FIRE

Phage Pharmers - question 6 of 8

Route used to deliver phages to CF lungs in the 2025 first-in-human randomized trial

Nebulized / inhaled

ROUND 11 - RAPID FIRE

Phage Pharmers - question 7 of 8

Chelator of divalent cations that can block phage adsorption



ROUND 11 - RAPID FIRE

Phage Pharmers - question 7 of 8

Chelator of divalent cations that can block phage adsorption

EDTA

ROUND 11 - RAPID FIRE

Phage Pharmers - question 8 of 8

Phages that kill the host and never lysogenize are called

ROUND 11 - RAPID FIRE

Phage Pharmers - question 8 of 8

Phages that kill the host and never lysogenize are called

Virulent (strictly lytic) phages

ROUND 11 - RAPID FIRE

Virion Architects - question 1 of 8

Tubulin-like phage protein that forms the spindle of jumbo phages



ROUND 11 - RAPID FIRE

Virion Architects - question 1 of 8

Tubulin-like phage protein that forms the spindle of jumbo phages

PhuZ

ROUND 11 - RAPID FIRE

Virion Architects - question 2 of 8

Structural term for the elongated (non-spherical) head of T4



ROUND 11 - RAPID FIRE

Virion Architects - question 2 of 8

Structural term for the elongated (non-spherical) head of T4

Prolate icosahedral head

ROUND 11 - RAPID FIRE

Virion Architects - question 3 of 8

Capsid vertex through which DNA enters during assembly



ROUND 11 - RAPID FIRE

Virion Architects - question 3 of 8

Capsid vertex through which DNA enters during assembly

Portal vertex (portal protein)

ROUND 11 - RAPID FIRE

Virion Architects - question 4 of 8

ATP-driven enzyme that packs DNA into the procapsid



ROUND 11 - RAPID FIRE

Virion Architects - question 4 of 8

ATP-driven enzyme that packs DNA into the procapsid

Terminase (large subunit)

ROUND 11 - RAPID FIRE

Virion Architects - question 5 of 8

Morphotype with a long, non-contractile tail



ROUND 11 - RAPID FIRE

Virion Architects - question 5 of 8

Morphotype with a long, non-contractile tail

Siphovirus

ROUND 11 - RAPID FIRE

Virion Architects - question 6 of 8

Geometric term for a capsid with 20 triangular faces



ROUND 11 - RAPID FIRE

Virion Architects - question 6 of 8

Geometric term for a capsid with 20 triangular faces

Icosahedral

ROUND 11 - RAPID FIRE

Virion Architects - question 7 of 8

Hollow tube through which the genome crosses the envelope in tailed phages



ROUND 11 - RAPID FIRE

Virion Architects - question 7 of 8

Hollow tube through which the genome crosses the envelope in tailed phages

Tail tube

ROUND 11 - RAPID FIRE

Virion Architects - question 8 of 8

The extra lipid layer present in ·6 but absent in T4



ROUND 11 - RAPID FIRE

Virion Architects - question 8 of 8

The extra lipid layer present in ϕ6 but absent in T4

Envelope (membrane)

12

TIE-BREAKER ROUND

Correct +20 | Wrong -10 | 60 sec | passed +5

ROUND 12

The T4 genome is circularly permuted and terminally redundant. Which packaging strategy generates this architecture?

A Headful packaging from concatemers

B cos-site cleavage of concatemers

C Protein-primed replication

D Rolling-circle of monomers

ROUND 12

The T4 genome is circularly permuted and terminally redundant. Which packaging strategy generates this architecture?

A Headful packaging from concatemers

B cos-site cleavage of concatemers

C Protein-primed replication

D Rolling-circle of monomers

ROUND 12

DNA replication of phage ·29 is primed by

A Host DnaA

B A primase (DnaG)

C A terminal protein

D A ribozyme

ROUND 12

DNA replication of phage ϕ29 is primed by

A Host DnaA

B A primase (DnaG)

C A terminal protein

D A ribozyme

ROUND 12

In the lambda decision circuit, high MOI together with a poor physiological host state favours

A Pseudolysogeny

B Lysis

C Superinfection

D Lysogeny (stabilized cII)

ROUND 12

In the lambda decision circuit, high MOI together with a poor physiological host state favours

A Pseudolysogeny

B Lysis

C Superinfection

D Lysogeny (stabilized cII)

ROUND 12

T4 glucosylates its 5-hydroxymethylcytosine DNA mainly to

A Protect the genome from host restriction endonucleases

B Aid adsorption

C Assist capsid assembly

D Speed up replication

ROUND 12

T4 glucosylates its 5-hydroxymethylcytosine DNA mainly to

A Protect the genome from host restriction endonucleases

B Aid adsorption

C Assist capsid assembly

D Speed up replication

ROUND 12

Which anti-phage system signals through a cyclic oligonucleotide (cGAMP) to activate a nuclease effector?

A Retrons

B Septu

C Gabija

D CBASS

ROUND 12

Which anti-phage system signals through a cyclic oligonucleotide (cGAMP) to activate a nuclease effector?

A Retrons

B Septu

C Gabija

D CBASS

ROUND 12

The lambda Red recombineering toolkit consists of the proteins

A EcoRI, EcoRV, ligase

B RecA, RecB, RecC

C Exo, Beta, Gam

D Int, Xis, Gam

ROUND 12

The lambda Red recombineering toolkit consists of the proteins

A EcoRI, EcoRV, ligase

B RecA, RecB, RecC

C Exo, Beta, Gam

D Int, Xis, Gam

ROUND 12

In the 2025 first-in-human randomized trial (BX004-A), the phage cocktail reached the CF lung by

A Intravenous infusion

B Nebulization

C Topical gel

D Oral capsules

ROUND 12

In the 2025 first-in-human randomized trial (BX004-A), the phage cocktail reached the CF lung by

A Intravenous infusion

B Nebulization

C Topical gel

D Oral capsules

ROUND 12

After a strong phage-driven selective sweep in a bacterial community, which metric is expected to DECREASE?

A Phage titer

B Shannon diversity index

C Burst size

D MOI

ROUND 12

After a strong phage-driven selective sweep in a bacterial community, which metric is expected to DECREASE?

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B Shannon diversity index

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13

EMOJI REBUS & PHAGE PERSONALS

Correct +10 | Wrong 0 | 30 sec

ROUND 13

FUN ROUND

[virus][scissors][DNA][book] · decode the rebus

A Restriction·modification

B CRISPR-Cas adaptive immunity

C Abortive infection

D Generalized transduction

ROUND 13

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FUN ROUND

[sleeping][UV sun][lightning][virus] · decode the rebus

A Prophage induction

B Lysogeny establishment

C Superinfection exclusion

D Pseudolysogeny

ROUND 13

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ROUND 13

FUN ROUND

[dynamite][city][virus] · decode the rebus

A Endolysin

B Spanin

C Depolymerase

D Integrase

ROUND 13

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[dynamite][city][virus] · decode the rebus

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ROUND 13

FUN ROUND

[box][DNA][balloon] · decode the rebus

A cos-site packaging

B Protein-primed replication

C Rolling-circle replication

D Headful (pac) packaging

ROUND 13

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[box][DNA][balloon] · decode the rebus

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ROUND 13

FUN ROUND

[detective][crab][test tube] · decode the rebus

A Plaque assay

B LAL endotoxin assay

C PFGE

D ELISA

ROUND 13 FUN ROUND

[detective][crab][test tube] · decode the rebus

A Plaque assay

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ROUND 13

FUN ROUND

[ice][microscope][virus] · decode the rebus

A Cryo-EM

B Negative-stain TEM

C Atomic force microscopy

D Phase-contrast light microscopy

ROUND 13

FUN ROUND

[ice][microscope][virus] · decode the rebus

A Cryo-EM

B Negative-stain TEM

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D Phase-contrast light microscopy

ROUND 13

FUN ROUND

[theatre masks][hand][DNA] · decode the rebus

A Phage typing

B Viromics

C Phage display

D Reassortment

ROUND 13 FUN ROUND

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ROUND 13

FUN ROUND

Phage personal ad: ·Seeking F-pilus-positive partner; ssDNA exchange welcome; no lysis required · I prefer a long, chronic togetherness.· Who wrote it?

A Phage T4

B Phage lambda

C Phage ·6

D Phage M13

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ROUND 13

FUN ROUND

AUDIENCE

Which set best describes a TEMPERATE phage?

A [sleeping][DNA][handshake]

B [boom][virus][skull]

C [dynamite][fire][skull]

D [runner][virus][timer]

ROUND 13

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ROUND 13

FUN ROUND

AUDIENCE

Personal ad: ·High-pressure type (~50 atm). I contract on first contact and deliver my DNA fast. No commitment issues · strictly lytic.· Identify the advertiser.

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C ·X174

D MS2

ROUND 13

FUN ROUND

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14

LYTIC LAUGHS

Correct +10 | Wrong 0 | 30 sec

ROUND 14

Why did the phage break up with the pilus?

A ·No chemistry in the periplasm.·

B ·It was too contractile.·

C ·Long-distance lysis never works.·

D ·The relationship was only surface-level attachment.·

ROUND 14

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ROUND 14

What is a bacteriophage's favourite genre of music?

A Ambient · like the eclipse period

B Smooth jazz for slow adsorption

C Opera, because it's always dramatic lysis

D Lytic metal · full of breakdowns

ROUND 14

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ROUND 14

Why don't phages use dating apps?

A No signal inside the gut

B They only ever swipe right on one specific host

C Too many matches cause lysis from without

D Their profiles are only ~50 kb

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ROUND 14

Where does a phage see itself in 30 minutes?

A ·In a preprint.·

B ·At a plaquelytics retreat.·

C ·As ~200 copies of myself inside one very surprised bacterium.·

D ·Integrated, settled and dormant.·

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ROUND 14

Why was the temperate phage terrible at commitment?

A It kept integrating into relationships and going dormant

B It lysed every relationship too fast

C It had no receptors for feelings

D It was too virulent to care

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What did the depolymerase say arriving at the biofilm party?

A ·I brought capsules for the road.·

B ·Relax · I'll break down the matrix and free everyone.·

C ·More EPS for everyone!·

D ·Who invited the antibiotics?·

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Why do phages make bad stand-up comedians?

A Their delivery is always lytic · they kill the room

B They are too small for the microphone

C They only infect the hecklers

D They have no tail for timing

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ROUND 14

What is a jumbo phage's excuse at the gym?

A ·I skip tail day.·

B ·My sheath is already too tight.·

C ·I carry a 500-kb genome · that's enough baggage.·

D ·No capsid space left.·

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Thank you for your answers!

See the scoreboard for final standings